Lab Report - Audio Amplifier (Stereo)

Aim:-

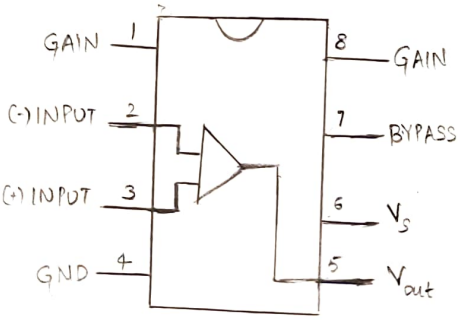
To construct a stereo audio amplifier circuit and demonstrate its working.

Components :-

1) LM386 - Low voltage audio power amplifier IC (x2)
2) Audio Jack (x1)
3) 1k Potentiometer (x2)
4) 7 - 10W Speaker (x2)
5) 10Ω Resistor (x2)
6) 4-12 V DC Input
7) 10μF Capacitor (x6)
8) 4.7nF Capacitor (x2)
9) 220μF, 16V Capacitor (x2)
10) 100μF Capacitor (x2)
11) 104μF Capacitor (x2)

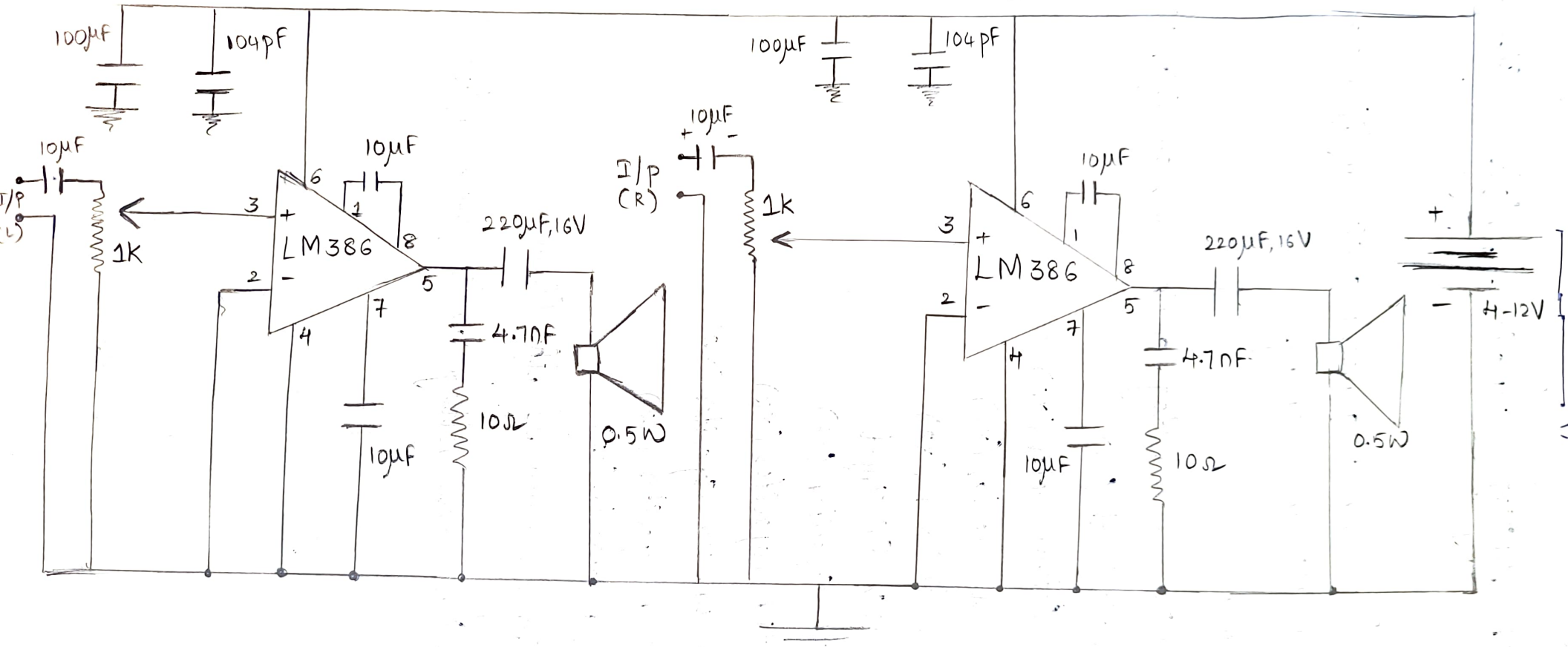
Working Procedure of the Circuit :-

1. LM386 - Low voltage audio power amplifier IC



LM386 Pin Diagram

Circuit Diagram :-



Audio Amplifier (Stereo)

⇒ LM386 is a low voltage audio amplifier with gain range 20 - 200. Without using external component, gain is 20 by default but can be increased to 200 by using resistor and/or capacitor between PIN 1 and 8 (GAIN PINS). It has a supply range of 4 - 12V.

(i) PIN 1 and 8 :-

These are GAIN CONTROL pins with an internal 1.35 kΩ resistor, that sets the gain to 20. If a 10μF capacitor is connected across PIN 1 and 8, bypassing the resistor, the gain becomes maximum i.e, 200. If a resistor is connected in series with capacitor, gain can be adjusted between 20 and 200.

(ii) PIN 2 and 3 :-

These are INPUT pins where PIN 2 is negative input terminal, connected to graund while PIN 3 is positive input terminal connected to sound signal. Here, it is connected to an audio jack with a 1k potentiometer that acts as a volume control knob. A 10μF capacitor is also used along with potentiometer to remove DC component of input signal and only feed the audio (AC component) into IC.

(iii) PIN 4 and 6 :-

These are Power supply pins where PIN 4 is a ground pin and PIN 6 is connected to V_s (Input voltage 4 - 12V). with 100μF and 104pF connected in parallel between V_s and ground.

(iv) PIN 5 :-

This is an Output pin where we get amplified sound signal as an output. This signal consists of AC and DC components, but DC cannot be fed to speaker. So, 220μF capacitor is used to remove this DC component. Also, a RC filter circuit made of 4.7nF & 10Ω resistor are used to remove any sudden high frequency oscillations/noise.

(v) Pin 7 :-

This is a Bypass terminal which can be left open or grounded using capacitor for stability.

⇒ 0.5 W Speaker :-

Outputs the amplified audio signal which is connected across 220μF capacitor and ground.

⇒ Audio Jack :-

Takes input sound from phone / laptop and inputs into PIN 5 of LM386 through potentiometer.

⇒ 1k Variable Resistor :-

Adjusts the magnitude of the input audio signal. It is used to control volume of the output audio signal.

Results :-

1. For an audio input signal of frequency 528 Hz,

$V_s = 0.055\,V$; $V_o = 1.113\,V$ without connecting 10μF b/w 1 & 8.

$$\therefore A = \frac{1.113}{0.055} \simeq 20$$

2. When 10μF capacitor is connected across PIN '1' & '8' :-

$V_s = 0.055\,V$; $V_o = 11.15\,V$

$$\therefore A' = \frac{11.15}{0.055} \simeq 200$$

Conclusion :-

The circuit successfully amplified the input audio signal using LM386. Various adjustments were made to minimise noise and produce a clear output. This circuit has various applications such as radios, intercoms etc.,

Precautions :-

1. Keep the input voltage between 4-12 V as the LM386 IC is designed to work in that range.
2. Take care of the polarity of the electrolytic capacitors while connecting them. Ensure that negative end is connected to ground and positive end to V_s.